A Project Report

on

MEDTWIN: REALISTIC AND PRIVATE SYNTHETIC PATIENT DATA WITH MISSING VALUES

carried out as part of the ***Project*** (Subject Code) *Submitted*

by

Pranav Ladha (23FE10CDS00455)
Devisha Jain (23FE10CDS00493)
Nandini Nahata (23FE10CDS00376)
Jiya Gakhar (23FE10CDS00480)

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MANIPAL UNIVERSITY
JAIPUR

School of Computer Science and Engineering,

MANIPAL UNIVERSITY JAIPUR
RAJASTHAN, INDIA

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CERTIFICATE

This is to certify that the project entitled **"MedTwin: Realistic and Private Synthetic Patient Data with Missing Values"** has been carried out by Pranav Ladha (23FE10CDS00455), Devisha Jain (23FE10CDS00493), Nandini Nahata (23FE10CDS00376) and Jiya Gakhar (23FE10CDS00480), students of the Department of Data Science and Engineering, Manipal University Jaipur, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology**. The work presented in this report is a bonafide record of the work done by them under my guidance and supervision during [start month, year] to [end month, year].

Dr. Chirag Joshi
Assistant Professor (Selection Grade)
Department of Data Science & Engineering
Manipal University Jaipur

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Pranav Ladha
Devisha Jain
Nandini Nahata
Jiya Gakhar

ABSTRACT

Patient data is essential for building medical machine-learning models, but privacy laws and hospital rules rightly restrict who may access it. Synthetic data — artificial patient records produced by a generative model that has learned the patterns of real records — is a promising way to share such data without exposing real people. Real clinical data, however, is incomplete in structured ways: whether a test was performed often depends on the hospital that recorded it. Most synthetic-data tools fill these gaps before learning, so the data they produce has no gaps at all, which makes it unrealistic and easy to tell apart from real records.

This project presents **MedTwin**, a conditional Wasserstein generative adversarial network with gradient penalty (WGAN-GP) that treats "not measured" as part of the record. For every column that can be missing it learns a yes/no measurement flag; a separate flag reproduces values that are exactly zero; and every synthetic patient is generated for a specific hospital, so that each hospital's measurement habits are learned. We also propose two new evaluation tools: missing-value fidelity scores, which measure whether the gaps in synthetic data are realistic per column, per hospital and as whole patterns, and a missing-value privacy audit, which tests whether keeping realistic gaps makes patients easier to identify.

MedTwin was compared with three established methods (CTGAN, TVAE and TabDDPM) on the four-hospital UCI Heart Disease data (920 patients) and the Pima Indians Diabetes data (768 patients), over five repeated experiments, always testing on real patients that no model had seen. MedTwin produced the most realistic records: a classifier could tell its heart-disease patients from real ones only 61.5% of the time, against 74.9–92.8% for the other methods (50% is ideal). A disease predictor trained only on its synthetic data kept 97.5% of the performance of one trained on real data, tied with the best other method. Its missing-value patterns were almost as close to the real data as a second sample of real patients (pattern distance 0.174 against 0.152), and it was the only method that placed gaps at the right hospitals. Privacy attacks found no sign of memorisation, whereas TabDDPM clearly memorised its training patients on the diabetes data. Finally, we found a small but consistent privacy cost of keeping realistic gaps for patients with unusual missing-value patterns — a new finding that we report openly.

Keywords: synthetic data, generative adversarial networks, missing data, electronic health records, privacy evaluation, deep learning

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
AUC	Area Under the (ROC) Curve
CSV	Comma-Separated Values (file format)
CTGAN	Conditional Tabular Generative Adversarial Network
DCR	Distance to Closest Record
EHR	Electronic Health Record
GAN	Generative Adversarial Network
MIA	Membership Inference Attack
ML	Machine Learning
MUJ	Manipal University Jaipur
RF	Random Forest
SD	Standard Deviation
TabDDPM	Tabular Denoising Diffusion Probabilistic Model
TRTR	Train on Real, Test on Real
TSTR	Train on Synthetic, Test on Real
TVAE	Tabular Variational Autoencoder
TVD	Total Variation Distance
UCI	University of California, Irvine (Machine Learning Repository)
VA	Veterans Affairs (VA Long Beach hospital)
WGAN-GP	Wasserstein GAN with Gradient Penalty
YAML	YAML Ain't Markup Language (settings file format)

CHAPTER 1
INTRODUCTION

1.1 Introduction to Work Done / Motivation (Overview, Applications & Advantages)

Overview

Machine learning is increasingly used in healthcare to predict diseases, support doctors in diagnosis and plan treatment. To build such models, researchers need large amounts of real patient data. This data is highly sensitive, and laws such as HIPAA (USA) and GDPR (Europe), together with hospital ethics rules, rightly restrict who can access it. As a result, students, researchers and smaller organisations often cannot obtain the data they need, and useful work is slowed down or never started.

Synthetic data offers a way around this problem [1, 2]. A generative model learns the statistical patterns of real patient records and then produces new, artificial records that follow the same patterns but do not belong to any real person. If the synthetic data is realistic, useful and private, it can be shared instead of the real data — for research, teaching, testing software and developing new models.

In this project we built **MedTwin**, a generative AI system that creates synthetic patient records from real hospital data. MedTwin is a conditional Wasserstein generative adversarial network with gradient penalty (WGAN-GP) [3, 4, 5]. It was tested on two public medical datasets — the four-hospital UCI Heart Disease data [6, 7] and the Pima Indians Diabetes data [8] — against three well-known synthetic-data methods.

Motivation

Real clinical tables are not complete, and the gaps are not random. Whether a test was done depends on the hospital, the doctor and the patient. In the heart disease data used in this project, for example:

- every patient from Switzerland has a cholesterol value of 0, which really means “not measured” (100% missing);
- the number of major blood vessels seen on a scan is missing for 96–99% of patients at three of the four hospitals, but almost never at Cleveland;
- resting blood pressure is missing for 28% of patients at VA Long Beach but for almost nobody elsewhere.

Most synthetic-data tools cannot handle missing values, so the usual practice is to fill each gap with a typical value (such as the median) before training. The synthetic data then has no gaps at all, which is not how real hospital data looks, and the filled-in values are inventions. A synthetic version of a multi-hospital dataset should reproduce *which* tests were done *where*. This observation is the main motivation of this project.

Applications

- **Data sharing for research:** hospitals can share realistic synthetic versions of their data with researchers without exposing real patients.
- **Education and training:** students can practise building medical machine-learning models on realistic data that contains real-world problems such as missing values.
- **Software and model testing:** developers can test data pipelines and prediction models on data that behaves like real hospital data, including its gaps.
- **Multi-hospital collaboration:** because MedTwin generates patients for a chosen hospital, it can represent the different measurement practices of several sites.
- **Benchmarking:** the evaluation tools developed here can be used to compare any synthetic-data generator on realism, usefulness, privacy and missing-value quality.

Advantages

- Synthetic records do not belong to real patients, reducing privacy risk when data is shared.
- Missing values are generated realistically instead of being invented, so the synthetic data keeps the structure of real data.
- The system is configured with a settings file, so a new dataset can be used without changing the code.
- Every result is tested on real patients the model never saw, repeated five times, and compared with paired statistics, so the conclusions are reliable.

1.2 Project Statement / Objectives of the Project

Problem statement. Design, implement and rigorously evaluate a generative AI system that produces realistic, useful and private synthetic patient records from incomplete, multi-hospital clinical data, keeping the real pattern of missing values; and develop methods to measure how realistic the missing values in synthetic data are and whether keeping them creates a privacy risk.

The main objectives of the project are:

1. **Build a missing-value-aware generator (MedTwin)** that learns which values are measured, generates each patient for a specific hospital, and reproduces values that are exactly zero.
2. **Make the system general**, so that any tabular medical dataset can be used by writing a settings file, and demonstrate this on two datasets.
3. **Evaluate fairly** on real patients held out from training, with five repeated experiments and paired statistical tests.
4. **Compare with established methods** — CTGAN, TVAE and TabDDPM — under identical conditions.
5. **Propose and apply missing-value fidelity scores** that measure how realistic the gaps in synthetic data are, per column, per hospital and as whole patterns.
6. **Audit the privacy cost of missing values** by testing whether keeping realistic gaps makes patients easier to identify.
7. **Explain each component's contribution** through ablation studies and check robustness through a sensitivity study.

1.3 Organization of Report

The rest of this report is organised as follows:

1. **Chapter 1 – Introduction:** the motivation, applications and advantages of the work, and the objectives of the project.
2. **Chapter 2 – Background Material:** the concepts and theory used (synthetic data, GANs, missing data, evaluation and privacy), the technologies involved, and a survey of related work.
3. **Chapter 3 – Methodology:** the datasets, data preparation, model design, training, evaluation protocol and the new missing-value measures, with block diagrams of the system.
4. **Chapter 4 – Implementation:** the software modules, the interactive dashboard prototype and the testing carried out.
5. **Chapter 5 – Results and Analysis:** comparison with other methods on both datasets, missing-value fidelity, the privacy audit, ablation and sensitivity studies, and challenges and learnings.
6. **Chapter 6 – Conclusions and Future Scope:** the main conclusions and recommended directions for future work.

7. **References and Annexures:** the sources cited (IEEE style), and supporting material such as sample synthetic records, code snippets and instructions to run the project.

CHAPTER 2
BACKGROUND MATERIAL

2.1 Conceptual Overview (Concepts / Theory Used)

1. Synthetic data

Synthetic data is artificial data produced by a model that has learned the patterns of real data. A good synthetic dataset must pass three tests at the same time [1, 19]: it must be **realistic** (it looks like real data, both column by column and in how columns relate), **useful** (models trained on it work on real data) and **private** (it does not reveal which real people were used to make it).

2. Generative Adversarial Networks (GANs)

A GAN [3] consists of two neural networks trained against each other. The **generator** turns random noise into fake records; the **discriminator** tries to tell fake records from real ones. As each network improves, the generator learns to produce records that are increasingly hard to distinguish from real data.

3. Wasserstein GAN with Gradient Penalty (WGAN-GP)

Standard GANs are often unstable and can collapse to producing only a few kinds of record. The Wasserstein GAN [4] replaces the discriminator's real/fake decision with a **critic** that outputs a realness score, estimating the Wasserstein distance between real and fake data. The gradient-penalty version [5] keeps the critic smooth by penalising its gradients, which makes training much more stable. The critic loss is:

$$L_D = \mathrm{E}[D(\text{fake}, c)] - \mathrm{E}[D(\text{real}, c)] + \lambda \cdot \mathrm{E}[(\|\nabla D(\text{mixed}, c)\| - 1)^2], \quad \text{with } \lambda = 10.$$

4. Conditional generation

In a conditional GAN, both networks receive extra information — the **condition**. In MedTwin the condition is the hospital: the generator creates a patient *for* a given hospital and the critic judges whether the patient is realistic *for that hospital*. This allows the model to learn each hospital's measurement habits.

5. Gumbel-softmax for categories

Categorical columns (such as chest pain type) must take exactly one value. The Gumbel-softmax technique [20] lets a neural network produce a smooth, trainable approximation of a one-hot choice during training and an exact one-hot choice when sampling.

6. Missing data

Missing values can arise for many reasons. In multi-hospital data they mostly reflect **site practice**: a hospital may simply not perform a test. Common tools fill missing values with a typical value (imputation) [21] or learn a separate model of which values are missing [22]. For synthetic data, imputation removes the gaps altogether, so the synthetic records no longer show which tests were done.

7. Evaluating synthetic data

- **Fidelity:** how closely the distribution of each column matches the real data (here measured by histogram overlap).
- **Detectability:** the accuracy of a classifier trained to tell real records from synthetic ones; 50% means they cannot be told apart. This measures realism, not privacy [23].
- **TSTR (Train on Synthetic, Test on Real)** [24]: a disease predictor is trained on synthetic data and tested on real patients; its AUC is compared with the same predictor trained on real data (TRTR).
- **Distance to Closest Record (DCR):** checks whether synthetic records are suspiciously close to the patients used for training.
- **Membership Inference Attack (MIA)** [25, 26]: an attacker tries to tell whether a given real patient was in the training data; an AUC of 0.5 means the attack fails.

Passing such attacks is evidence, not a guarantee, of privacy [27].

8. Statistical testing and ablation

Because results vary with random choices, each experiment is repeated with five seeds. Methods are compared seed by seed using a paired t-based 95% confidence interval; a difference is called **clear** only when the interval does not include zero. An **ablation study** switches off one part of the system at a time to measure how much that part contributes.

2.2 Technologies Involved

The project was implemented entirely in Python using open-source libraries. Table 2.1 lists the main technologies and what each was used for.

Table 2.1: Technologies and tools used

Area	Technology / version	Used for
Programming language	Python 3.9	All code
Deep learning	PyTorch 2.8.0 [9]	Generator, critic and training
Data handling	pandas 2.3.3, NumPy 2.0.2	Loading, cleaning and encoding data

Area	Technology / version	Used for
Machine learning	scikit-learn 1.6.1 [10]	Random forests, nearest neighbours, AUC, scaling
Comparison methods	ctgan 0.12.1 (CTGAN, TVAE); TabDDPM written in this project	Baselines
Statistics	SciPy	Paired t-tests and confidence intervals
Visualisation	matplotlib 3.9.4, seaborn 0.13.2	Figures
Dashboard	Streamlit 1.50.0	Interactive web prototype
Configuration	PyYAML 6.0.3	Dataset settings files
Testing	pytest	31 automated tests
Version control	Git	History of all changes
Documents	LaTeX (Springer LNCS), Tectonic	Research paper

Random forests [28] were used for the detectability, utility and privacy tests, and the Adam optimiser [29] for training the networks. The baselines CTGAN and TVAE [30] come from the open-source ctgan library, and TabDDPM [31] was re-implemented in this project following its published design.

2.3 Literature Survey

Early work on synthetic health records used GANs for binary and count features (medGAN [32]). CTGAN [30] introduced a conditional GAN for mixed tables of numbers and categories, and the same work proposed TVAE, a variational autoencoder for tables. CTAB-GAN [33] improved the handling of mixed and long-tailed variables. Diffusion models now lead many tabular benchmarks: TabDDPM [31] and TabSyn [34]. EHR-Safe [35] addresses the complexity of real electronic health records. Large benchmark studies [2, 19, 36] compare many generators on realism, usefulness and privacy, and recent work shows that tabular diffusion models can memorise their training records [37].

Several recent studies generated synthetic versions of the same or similar data. Table 2.2 compares them with this project. Exact number-for-number comparison is usually not possible because each study uses different data, splits and measures.

Table 2.2: Comparison with recent related work

Work	What they did	Their result	How this project compares
Wan Abu Bakar et al., 2025 [11]	CTGAN on Cleveland heart data (303 patients); gaps filled with the most common value; single run	56% accuracy training on synthetic, testing on real (5-class)	We use all 4 hospitals, unseen test patients and 5 repeats; MedTwin clearly beats our own CTGAN baseline

Work	What they did	Their result	How this project compares
Subedi et al., 2025 [12]	CTGAN on Pima diabetes	Random forest accuracy 82% (real) → 63% (synthetic)	MedTwin keeps 98% of real-data performance (AUC ratio)
Alqulaity & Yang, 2024 [13]	Enhanced conditional GAN, 218 cardiovascular patients, one hospital	Distribution statistics only; no utility or privacy test	We test realism, usefulness and privacy on unseen patients
Lazzaro et al., 2026 [14]	Bayesian networks on UCI heart sites, trained on Cleveland only	Data augmentation results	We model all 4 hospitals, including the very incomplete ones
HealthGen [15], MissDiff [16], Wang et al. [17], IGAMT [18]	Generators that keep missing values (time series, diffusion, plug-in, private EHR)	Show that keeping missingness matters	Same goal; we add hospital-conditioned gaps, scores for gap realism and the first audit of their privacy cost

2.4 Research Gap

From the survey we identified the following gaps, which this project addresses:

1. Studies on the UCI heart and Pima data usually fill missing values, use a single run, test on patients the generator saw, and do not measure privacy.
2. Methods that keep missing values exist, but there is no standard way to measure whether the generated gaps are realistic, particularly per hospital.
3. No study has tested whether keeping realistic missing values creates a privacy risk for patients with unusual gaps.

CHAPTER 3
METHODOLOGY

3.1 Detailed Methodology that will be Adopted

The project was carried out in eight stages, summarised in the block diagram of Fig. 3.1 (Section 3.2). Each stage is described below.

Step 1: Data collection and understanding

Two public datasets were used. The **UCI Heart Disease** data [6, 7] combines patients from four hospitals; the **Pima Indians Diabetes** data [8] was obtained from OpenML (dataset 37) [38]. Table 3.1 summarises both datasets.

Table 3.1: Summary of the datasets

	Heart disease (UCI)	**Pima diabetes**
Patients	920	768
Columns used	7 numbers + 7 categories + hospital	9 numbers
Hospitals	Cleveland (304), Hungary (293), VA Long Beach (200), Switzerland (123)	1 (one clinic)
Prediction target	heart disease (severity > 0): 55.3% yes	diabetes: 34.9% yes
Missing values	many; depend strongly on the hospital	zeros that mean “not measured”

Table 3.2 lists the heart disease columns and how often each is missing. Table 3.3 shows that the missing values depend strongly on the hospital — this is exactly the structure a realistic synthetic dataset must reproduce.

Table 3.2: Heart disease columns

Column	**Meaning**	**Type**	**Missing**
age	Age in years	number	0.0%
sex	Sex	category	0.0%
dataset	Hospital (site)	site	0.0%
cp	Chest pain type	category	0.0%
trestbps	Resting blood pressure (mmHg)	number	6.5%
chol	Serum cholesterol (mg/dl)	number	22.0%
fbs	Fasting blood sugar > 120 mg/dl	category	9.8%
restecg	Resting ECG result	category	0.2%
thalch	Maximum heart rate (bpm)	number	6.0%
exang	Chest pain during exercise	category	6.0%
oldpeak	ST depression on ECG	number	6.7%
slope	Slope of ST segment	category	33.6%

Column	Meaning	Type	Missing
ca	Major vessels seen on scan (0–3)	number	66.4%
thal	Thalassemia test result	category	52.8%
num	Disease severity (0 = none, 1–4)	number	0.0%

Table 3.3: Share of heart disease patients with each value not measured, by hospital

Hospital (patients)	chol	trestbps	ca	thal	slope	fbs
Cleveland (304)	0%	0%	2%	1%	0%	0%
Hungary (293)	8%	0%	99%	90%	65%	3%
Switzerland (123)	100%	2%	96%	42%	14%	61%
VA Long Beach (200)	28%	28%	99%	83%	51%	4%

For the diabetes data, a value of 0 for glucose, blood pressure, skin fold, insulin or BMI is physically impossible and means “not measured” (insulin is missing for 48.7% of patients). Zero pregnancies, however, is a real value. These rules are written in each dataset's settings file.

Step 2: Train / test split

For each experiment, 20% of patients are held out and never shown to any model. All usefulness and privacy tests are measured against these unseen patients. The split is stratified by the outcome so both parts have the same share of sick patients.

Step 3: Encoding records with their missing values

- **Zeros that mean missing** (from the settings file) are turned into blanks.
- **Missing-value flags:** every number column that is sometimes missing gets a 0/1 flag “was this measured?”. The blank value itself is replaced by the median as a placeholder that the flag marks as meaningless.
- **Zero flag:** 40% of heart patients have an ST depression of exactly zero; a separate 0/1 flag lets the network reproduce this spike.
- **Scaling and categories:** numbers are scaled to the range 0–1; each category column is one-hot encoded with “not measured” as an extra category.
- **Condition:** the hospital is not generated but given to the model as a one-hot condition.

Step 4: Model design and training

MedTwin is a conditional WGAN-GP. Its architecture is given in Table 3.4 and drawn in Fig. 3.2. In addition to the Wasserstein loss and gradient penalty, the generator is penalised

when the correlations between columns in its output differ from those in the real batch (correlation loss, weight $\alpha = 2$). Columns that are constant within a batch are skipped, because their correlation is undefined and would otherwise break training. Training settings are listed in Table 3.5.

Table 3.4: MedTwin model architecture

Part	Specification
Generator input	Random noise (128 numbers) + one-hot hospital condition
Generator layers	Linear 256 → LeakyReLU(0.2) → BatchNorm → Linear 512 → LeakyReLU → BatchNorm → Linear (output)
Generator outputs	Sigmoid for numbers and flags (0–1); Gumbel-softmax ($\tau = 0.2$) per categorical column
Critic layers	Linear 512 → LeakyReLU → Dropout 0.3 → Linear 256 → LeakyReLU → Dropout 0.3 → Linear 1
Critic input	Record + hospital condition, so realism is judged per hospital
Loss	Wasserstein loss + gradient penalty ($\lambda = 10$) + correlation penalty ($\alpha = 2$)

Table 3.5: Training settings

Setting	Value
Optimiser	Adam [29], learning rate 0.0001, betas (0.0, 0.9)
Epochs / batch size	600 epochs, batch size 64
Critic updates	5 critic updates per generator update
Records generated	2000 per run
Train / test split	80% / 20%, stratified by outcome
Repeats	5 seeds (0–4); each seed changes the split and the model's starting point

Step 5: Generation and decoding

To create synthetic data, a hospital is drawn for each patient in the same proportions as the real data, and the generator produces 2000 patients. Values are converted back to real units. If a patient's zero flag is set, the value becomes 0; if its measurement flag is set, the value is left blank (“not measured”).

Step 6: Cleaning

Generated values are rounded to the precision used in hospital records (for example, ST depression to 0.1) and checked against medically possible ranges. The ranges were chosen to include every real measured value, so no real patient would be removed.

Step 7: Evaluation protocol

All four methods receive the same data, the same missing-value flags, the same cleaning and the same tests. Each experiment is repeated with five seeds, and methods are compared seed by seed with paired 95% confidence intervals. Table 3.6 lists every measure used; those marked “new” were developed in this project.

Table 3.6: Evaluation measures

Measure	Question it answers	Best	Group
Fidelity (%)	Do single columns look real? (histogram overlap)	Higher	Realism
Detectability (%)	Can a classifier tell fake from real?	50%	Realism
Correlation gap	Do relationships between columns match?	0	Realism
TSTR AUC / AUC ratio	Does a predictor trained on fake data work on real patients?	Higher / 1.0	Usefulness
DCR (%)	Are fake records closer to training patients than to unseen ones?	50%	Privacy
MIA AUC	Can an attacker tell which patients were used for training?	0.5	Privacy
Rate gap (new)	Is each column missing as often as in real data?	0	Missing values
Site gap (new)	Is each column missing as often at each hospital?	0	Missing values
Pattern distance (new)	Do whole patterns of gaps match?	0	Missing values
Rare-pattern MIA (new)	Are patients with unusual gaps easier to identify?	0.5	Privacy of gaps
Unique-pattern leak (new)	Do one-of-a-kind gap patterns of training patients reappear?	0	Privacy of gaps

Step 8: New missing-value measures

A patient's **missingness pattern** is the list of which of their values are blank, together with their hospital. Using this idea we defined:

- **Rate gap:** the average difference, over columns, in the share of blank values between real and synthetic data.
- **Site gap:** the same difference computed for each hospital and column, averaged with hospital weights. It catches a model that gets overall rates right but puts gaps at the wrong hospital.
- **Pattern distance:** the total variation distance between the distributions of whole patterns, ½ Σ |P_real(p) − P_synthetic(p)|. It also captures which values go missing together.
- **Real-vs-real reference:** because these scores depend on sample size, the synthetic data is scored on samples the size of the held-out set, and the held-out real patients

are scored the same way. A synthetic score close to this reference is as good as a second sample of real patients.

- **Rare-pattern membership attack:** the distance-based membership attack is run separately on patients whose pattern is rare (held by at most 2% of patients) and on the rest.
- **Unique-pattern leak:** for patterns held by exactly one patient, the share that reappears in the synthetic data when that patient was in training, minus the share when they were held out.

Ablation and sensitivity design

To measure what each component contributes, MedTwin was retrained with one component switched off at a time: missing-value flags, zero flag, hospital conditioning and correlation loss. To check that results do not depend on lucky settings, the correlation weight (0.5 and 8 instead of 2) and training length (300 and 1200 instead of 600 epochs) were varied.

3.2 Circuit Layouts / Block Diagrams

Fig. 3.1 shows the complete MedTwin pipeline, from the real data and settings file to the evaluated synthetic data.

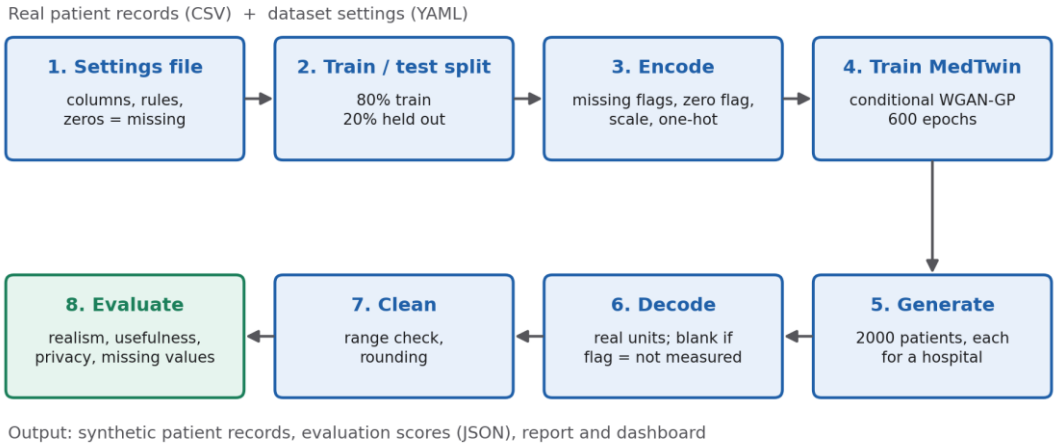


Figure 3.1: Block diagram of the MedTwin pipeline

Fig. 3.2 shows the conditional WGAN-GP. The generator receives noise and the hospital; its outputs are split into sigmoid outputs (numbers and flags) and Gumbel-softmax outputs (categories). The critic receives either a generated or a real record together with the same hospital and outputs a realness score.

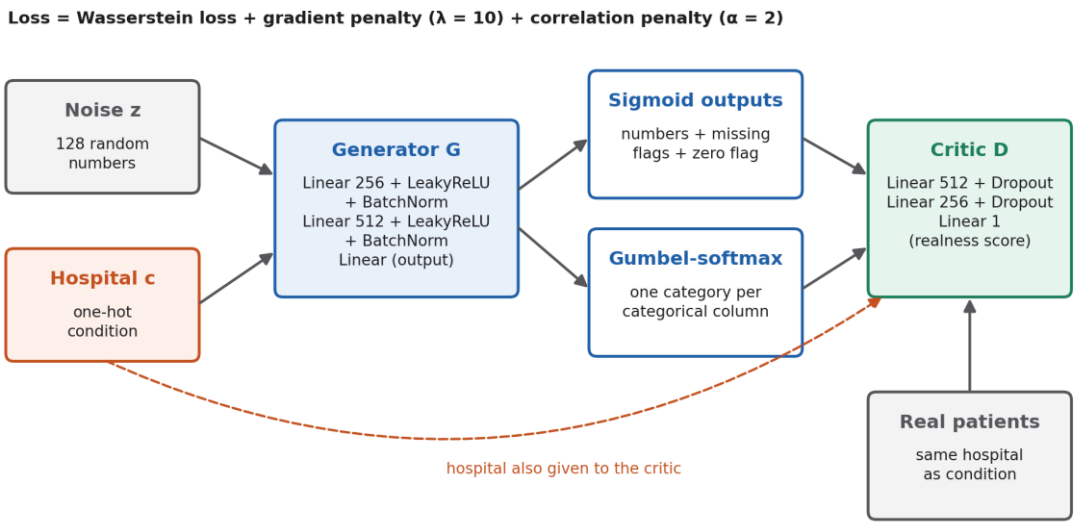


Figure 3.2: Architecture of the MedTwin conditional WGAN-GP

Fig. 3.3 shows the evaluation framework: the held-out patients are never seen by the generator and are used for all usefulness, privacy and missing-value tests.

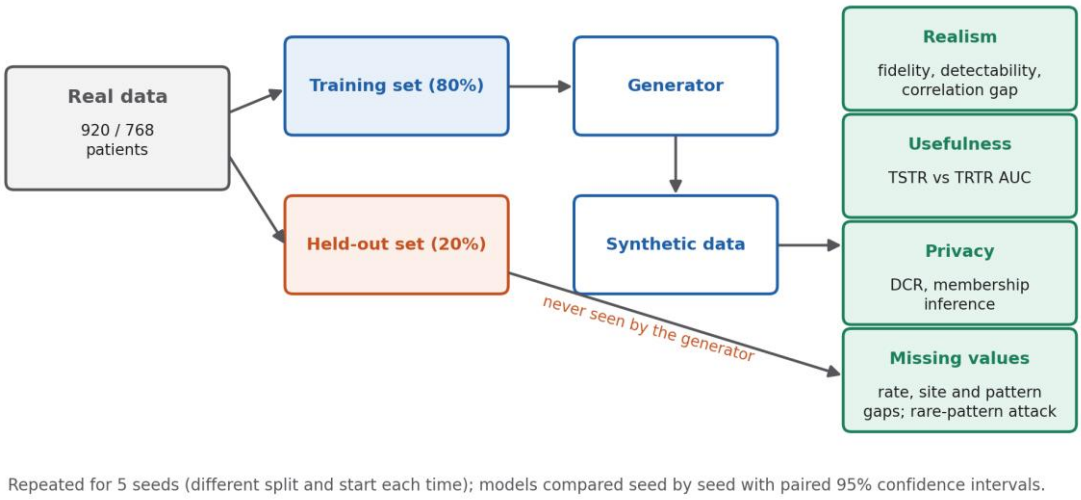


Figure 3.3: Evaluation framework

CHAPTER 4
IMPLEMENTATION

4.1 Modules

The system is organised into small, separately tested modules. Table 4.1 lists the main files and their purpose.

Table 4.1: Software modules

File / folder	Purpose
main.py	Runs the full pipeline once and saves synthetic data and scores
core/config.py	Loads a dataset's settings file (columns, rules, ranges)
core/ingestion.py	Missing-value handling, flags, scaling, encoding
core/generator.py	Generator, critic and WGAN-GP training loop
core/medtwin.py	Train-and-generate steps; ablation and sensitivity switches
core/medical_logic.py	Medical range check and rounding
core/evaluation.py	Usefulness, privacy and correlation tests on held-out patients
core/missingness.py	New missing-value fidelity scores and privacy attacks
research/baselines.py, tabddpm.py	The three comparison methods
research/multi_seed.py	Five-seed comparison with paired statistics
research/ablation.py	Ablation and sensitivity studies
research/visualizer.py, paper_tables.py	Figures and paper tables
app/dashboard.py	Interactive Streamlit dashboard
configs/*.yaml	One settings file per dataset
tests/	31 automated tests

1. Settings module

Everything specific to a dataset lives in a YAML settings file: which columns are numbers or categories, which column is the hospital, which zeros mean “not measured”, which columns have a spike at zero, the medically possible ranges, and the prediction target. Adding a new dataset means writing a new settings file, not changing code — this is how the diabetes data was added.

2. Data ingestion module

This module marks zero-coded values as missing, creates the missing-value and zero flags, fills blanks with placeholder medians, scales numbers to 0–1, one-hot encodes categories (with “not measured” as a category) and prepares the hospital condition. It also records the layout of the generator's outputs so that the model can be built automatically for any dataset.

3. Generator and training module

This module defines the conditional generator and critic (Table 3.4) and the WGAN-GP training loop with five critic updates per generator update, the gradient penalty and the correlation loss. A bug found during development — a batch with a constant column produced NaN values that corrupted the whole model — was fixed by skipping constant columns in the correlation loss.

4. Generation, cleaning and evaluation modules

After training, synthetic patients are generated for hospitals drawn in real proportions, decoded back to real units, blanked where the flags say "not measured", range-checked and rounded. The evaluation module then computes fidelity, detectability, TSTR utility, DCR, membership inference and the correlation gap against the held-out patients, and the missingness module computes the new missing-value scores and privacy attacks.

5. Baselines and experiment modules

The comparison methods were run through exactly the same pipeline (Table 4.2). Because none of them accepts missing values, they were given the same flag-and-placeholder encoding as MedTwin. Experiment runners repeat every comparison over five seeds, save each run's results and synthetic data, and can resume if interrupted. When all experiments were re-run after new measurements were added, all 400 previously saved results came out exactly the same, confirming that the experiments are reproducible.

Table 4.2: Methods compared and average training time (heart disease)

Method	Description	Training time
CTGAN [30]	Conditional tabular GAN; 300 epochs, batch 100	1.2 min
TVAE [30]	Variational autoencoder for tables; 300 epochs, batch 100	0.4 min
TabDDPM [31]	Diffusion model for tables (our re-implementation); 8000 steps	3.2 min
MedTwin (ours)	Conditional WGAN-GP with missing-value flags; 600 epochs	6.8 min

4.2 Prototype

A working prototype was built as an interactive web dashboard using Streamlit (Fig. 4.1). It loads the latest synthetic data and shows: the number of synthetic patients, statistical fidelity, an average clinical value and the detectability score; the held-out evaluation (TSTR utility, utility ratio, DCR and membership-inference AUC); interactive distribution and correlation comparisons between real and synthetic patients; and a data explorer in which

blank values are shown as “not measured”. Users can filter the synthetic patients by sex and age range and export the filtered records as a CSV file.

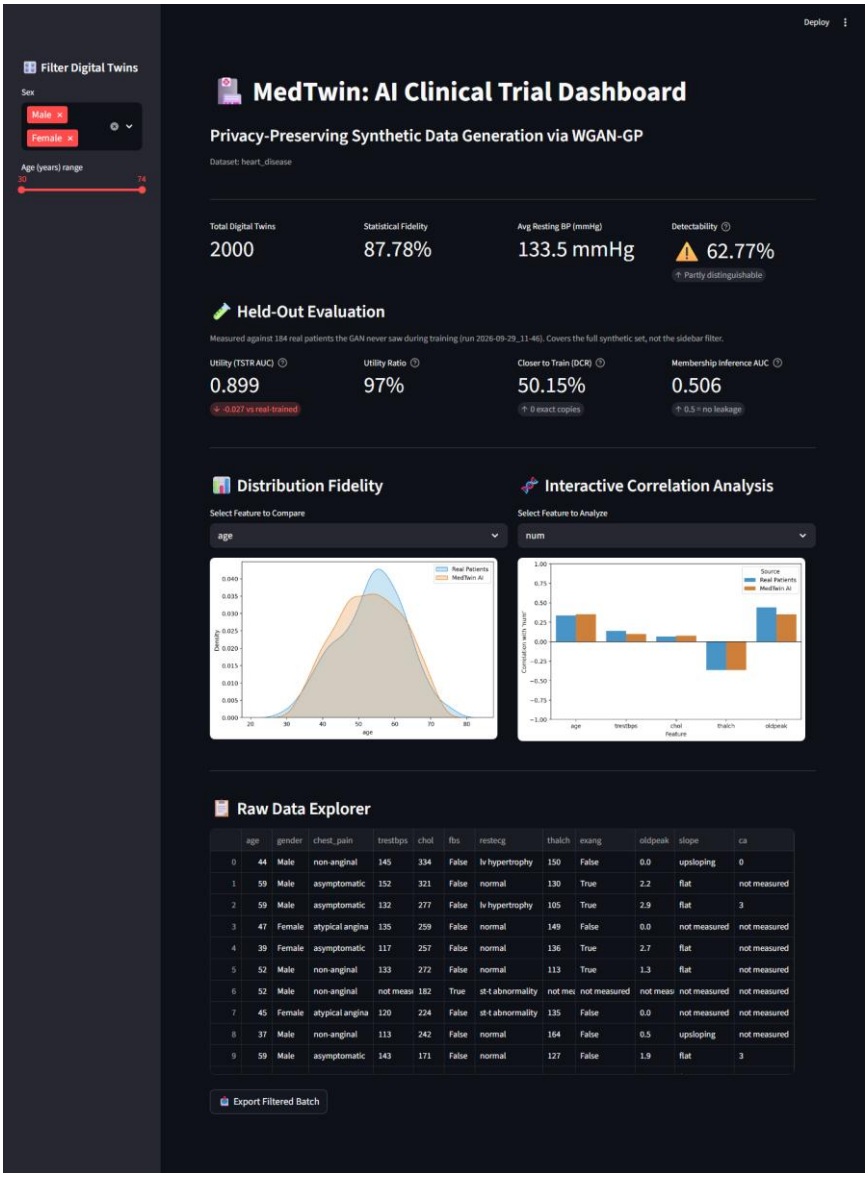


Figure 4.1: MedTwin interactive dashboard (heart disease data)

The pipeline can also be run from the command line: one command trains MedTwin, generates synthetic patients, runs all tests and prints a summary of fidelity, detectability, utility, privacy and missing-value scores. A single option switches to another dataset (Annexure C).

4.3 Testing

The project includes 31 automated tests (pytest) that run in under a minute with very short training. They check, among other things, that: the training matrix has no blanks and matches the generator's output layout; every generated category is exactly one value;

conditioning on a hospital produces patients only for that hospital; all generated values respect the medical ranges; real patients handed in as “synthetic” are caught by the privacy tests; data with every gap filled gets a bad missing-value score; every ablation variant trains and produces a valid report; and results are unchanged when synthetic data is saved to CSV and read back.

CHAPTER 5
RESULTS AND ANALYSIS

All results are averages over five repeated experiments ± standard deviation, measured on real patients that the models never saw. Arrows show what is better: ↑ higher, ↓ lower, → closer to the given value. A difference is called **clear** when its paired 95% confidence interval excludes zero.

5.1 Heart Disease: MedTwin Compared with Other Methods

Table 5.1: Heart disease results (mean ± SD over five seeds)

Method	Fidelity % (↑)	Detect. % (→50)	TSTR AUC (↑)	AUC ratio (→1)	DCR % (→50)	MIA AUC (→0.5)
MedTwin (ours)	87.5 ± 0.8	61.5 ± 1.3	0.863 ± 0.023	0.975 ± 0.032	52.9 ± 2.8	0.533 ± 0.030
TVAE	85.6 ± 1.1	79.5 ± 2.8	0.873 ± 0.022	0.987 ± 0.024	50.1 ± 3.4	0.531 ± 0.023
TabDDPM	86.1 ± 1.1	74.9 ± 2.8	0.820 ± 0.030	0.926 ± 0.022	52.7 ± 3.0	0.526 ± 0.033
CTGAN	74.2 ± 4.6	92.8 ± 1.6	0.696 ± 0.069	0.787 ± 0.082	51.1 ± 3.4	0.507 ± 0.020

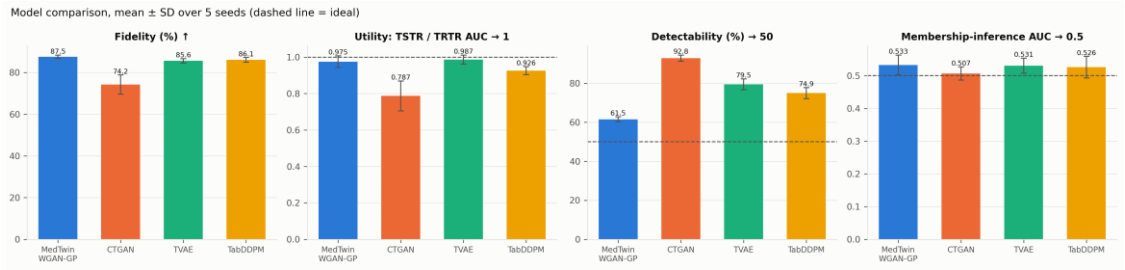


Figure 5.1: Heart disease comparison of the four methods (mean ± SD over five seeds; dashed lines show the ideal value)

Results:

- **Most realistic:** MedTwin has the highest fidelity (87.5%) and by far the lowest detectability (61.5%, against 74.9% for TabDDPM, 79.5% for TVAE and 92.8% for CTGAN). Both differences are clear against every method.
- **Just as useful:** the AUC ratio is 0.975, statistically tied with TVAE (0.987) and clearly better than CTGAN (0.787).
- **Private:** DCR and membership-inference AUC are close to their ideal values for all methods, and no method produced an exact copy of a training patient.

Analysis: generating each patient for a hospital and generating missing values, instead of filling them in, gives MedTwin records that a classifier finds much harder to tell apart from real patients, without losing usefulness. Detectability (61.5%) is still above the ideal 50%, so there is room for improvement.

5.2 Diabetes: The Same Picture

Table 5.2: Pima diabetes results (mean ± SD over five seeds)

Method	Fidelity % (↑)	Detect. % (→50)	TSTR AUC (↑)	AUC ratio (→1)	DCR % (→50)	MIA AUC (→0.5)
MedTwin (ours)	90.9 ± 0.8	64.0 ± 2.6	0.801 ± 0.030	0.983 ± 0.041	51.8 ± 1.8	0.494 ± 0.021
TVAE	88.0 ± 1.7	70.0 ± 1.4	0.806 ± 0.013	0.989 ± 0.028	54.0 ± 2.4	0.518 ± 0.024
TabDDPM	85.4 ± 1.3	55.9 ± 2.6	0.781 ± 0.050	0.957 ± 0.026	60.8 ± 4.9	0.803 ± 0.028
CTGAN	77.8 ± 3.3	86.3 ± 2.7	0.568 ± 0.062	0.697 ± 0.075	49.7 ± 7.7	0.505 ± 0.028

- MedTwin again has the best fidelity (90.9%) and ties for usefulness (AUC ratio 0.983).
- TabDDPM is the hardest to detect (55.9%) **but it memorised its training patients**: 60.8% of its records are closer to training patients than to unseen ones, and the membership attack reaches an AUC of 0.803. MedTwin shows no such leak (0.494). This agrees with recent findings that tabular diffusion models memorise [37], and shows why detectability alone must not be read as privacy.

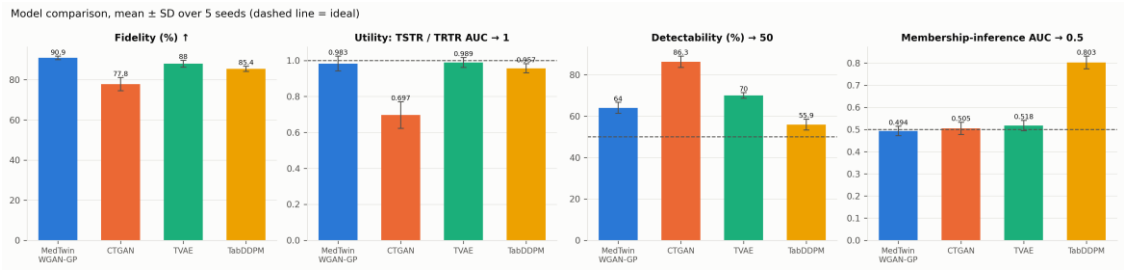


Figure 5.2: Pima diabetes comparison of the four methods

5.3 Are the Missing Values Realistic?

Table 5.3: Missing-value fidelity (mean ± SD; lower is better)

Data	Rate gap	Site gap	Pattern distance
Heart disease			
Two real samples (reference)	0.020	0.024	0.152
MedTwin (ours)	0.021 ± 0.002	0.022 ± 0.001	0.174 ± 0.006
TVAE	0.028 ± 0.005	0.035 ± 0.002	0.209 ± 0.009
TabDDPM	0.032 ± 0.005	0.101 ± 0.004	0.487 ± 0.028
CTGAN	0.059 ± 0.005	0.131 ± 0.006	0.683 ± 0.014
MedTwin without missing-value flags	0.118 ± 0.002	0.121 ± 0.002	0.693 ± 0.009
MedTwin without hospital conditioning	0.019 ± 0.002	0.028 ± 0.001	0.223 ± 0.012
Diabetes (one clinic, so no site gap)			
Two real samples (reference)	0.030	–	0.085
MedTwin (ours)	0.019 ± 0.002	–	0.063 ± 0.006

Data	Rate gap	Site gap	Pattern distance
TVAE	0.031 ± 0.005	–	0.118 ± 0.022
TabDDPM	0.029 ± 0.010	–	0.111 ± 0.013
CTGAN	0.089 ± 0.033	–	0.299 ± 0.032
MedTwin without missing-value flags	0.167 ± 0.006	–	0.486 ± 0.018

- MedTwin's pattern distance on heart disease (0.174) is close to the gap between two real samples (0.152) and clearly better than every other method in all five seeds (TVAE 0.209, TabDDPM 0.487, CTGAN 0.683).
- **Only MedTwin gets the hospital right:** its site gap (0.022) matches real data (0.024). The other methods were given the same flags and their overall missing rates are close to real, but they do not know which hospital a gap belongs to, so they put gaps in the wrong places (site gaps 0.035–0.131). A per-column check alone would not reveal this.
- Without the missing-value flags the pattern distance jumps to 0.693; without hospital conditioning the overall rates stay right but the site gap (0.028) and pattern distance (0.223) clearly worsen.
- On diabetes, MedTwin again has the lowest pattern distance (0.063; other methods 0.111–0.299).

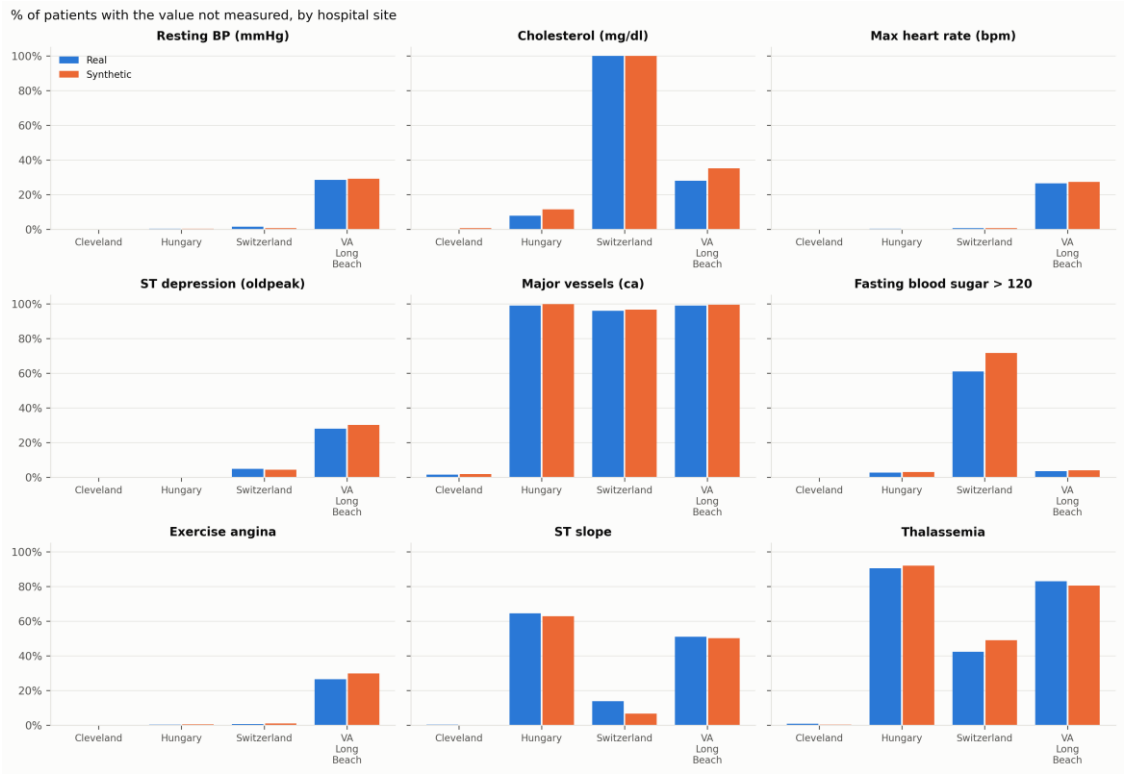


Figure 5.3: Share of heart disease patients with each value not measured, per hospital: real (blue) and MedTwin (orange)

Fig. 5.3 shows that MedTwin reproduces the hospital-specific structure: cholesterol missing for every Swiss patient, vessel counts missing almost everywhere except Cleveland, and thalassemia missing for most Hungarian and VA patients. The largest differences are for fasting blood sugar in Switzerland (61% real, 72% synthetic) and ST slope in Switzerland (14% real, 7% synthetic).

5.4 Does Keeping Missing Values Cost Privacy?

Table 5.4: Missing-value privacy audit, heart disease (mean ± SD; 0.5 or 0 is ideal)

Variant	MIA rare patterns	MIA common patterns	MIA all patients	Unique leak (points)
Full MedTwin	0.575 ± 0.046	0.542 ± 0.042	0.532 ± 0.030	-4.7 ± 25.3
MedTwin without missing-value flags	0.531 ± 0.049	0.516 ± 0.016	0.515 ± 0.027	1.9 ± 21.8

Results: on heart disease about 64 of the 368 attacked patients per test have a rare missingness pattern. For them, the membership attack reaches an AUC of **0.575** against MedTwin, compared with 0.531 when MedTwin is trained without the missing-value flags — higher in all five seeds. For patients with common patterns and over all patients the difference is not clear. Patterns unique to one training patient do not reappear more often than those of unseen patients, so the model is not copying patterns.

Analysis: keeping realistic gaps makes the data more realistic and leaves overall privacy unchanged, but patients with *unusual* gaps become slightly easier to single out. The effect is modest, but it concerns exactly the group a data owner would worry about. On diabetes only about 8 patients per test have rare patterns, too few for a conclusion.

5.5 What Each Component Contributes (Ablation)

Table 5.5: Ablation study (mean ± SD; * = clear difference from the full model)

Variant	Detect. % (→50)	Corr. gap (↓)	AUC ratio (→1)	MIA AUC (→0.5)
Heart disease				
Full MedTwin	62.2 ± 1.7	0.041 ± 0.007	0.974 ± 0.028	0.532 ± 0.030
without missing-value flags	84.2 ± 1.2 *	0.071 ± 0.007 *	0.989 ± 0.025	0.515 ± 0.027
without zero flag	74.9 ± 0.9 *	–	0.978 ± 0.032	0.539 ± 0.038
without hospital conditioning	65.6 ± 1.7 *	0.046 ± 0.005	0.975 ± 0.038	0.527 ± 0.025
without correlation loss	65.2 ± 2.4	0.075 ± 0.018 *	0.982 ± 0.011	0.525 ± 0.032
Diabetes				
Full MedTwin	64.0 ± 2.6	0.034 ± 0.007	0.983 ± 0.041	0.494 ± 0.021

Variant	Detect. % (→50)	Corr. gap (↓)	AUC ratio (→1)	MIA AUC (→0.5)
without missing-value flags	81.5 ± 1.6 *	0.042 ± 0.004 *	0.993 ± 0.032	0.524 ± 0.055
without correlation loss	63.8 ± 2.9	0.063 ± 0.008 *	0.964 ± 0.032	0.508 ± 0.029

- **Missing-value flags matter most:** without them detectability rises from 62.2% to 84.2% (heart) and from 64.0% to 81.5% (diabetes). Filling gaps with typical values makes synthetic patients easy to spot, consistent with the bias of "impute-then-generate" reported in [16].
- **Zero flag and hospital conditioning** also clearly help realism (74.9% and 65.6% without them).
- **Correlation loss** does not change realism or usefulness, but without it the correlation gap between numeric columns nearly doubles (0.041 → 0.075 heart; 0.034 → 0.063 diabetes).
- No component changes usefulness or overall privacy. The medical range check never removed a patient: it is a safety net.

5.6 Sensitivity to Settings

Table 5.6: Sensitivity of MedTwin to its main settings, heart disease (* = clear difference from the default)

Setting	Fidelity %	Detect. %	AUC ratio	DCR %	MIA AUC
Default (weight 2, 600 epochs)	87.2 ± 0.7	62.2 ± 1.7	0.974 ± 0.028	53.3 ± 2.0	0.532 ± 0.030
Correlation weight 0.5	87.0 ± 0.8	62.8 ± 2.8	0.983 ± 0.021	53.3 ± 2.1	0.531 ± 0.023
Correlation weight 8	87.9 ± 0.5	62.8 ± 2.3	0.989 ± 0.031	53.9 ± 2.6	0.525 ± 0.032
300 epochs	87.0 ± 0.5	63.6 ± 2.8	0.961 ± 0.034	52.2 ± 2.5	0.524 ± 0.035
1200 epochs	87.2 ± 0.5	63.0 ± 2.2	0.984 ± 0.017	55.3 ± 2.6 *	0.548 ± 0.024 *

Changing the correlation weight or training for 300 epochs makes no clear difference. Training twice as long (1200 epochs) gives the same realism and usefulness but starts to memorise: DCR rises from 53.3% to 55.3% and MIA AUC from 0.532 to 0.548. The results therefore do not depend on lucky settings, and 600 epochs is a good stopping point.

5.7 Quality of the Generated Values

Figs. 5.4 and 5.5 compare the distributions of the number columns and the correlations between them for real and synthetic heart disease patients. The shapes and the main relationships — such as maximum heart rate falling with age — are preserved.

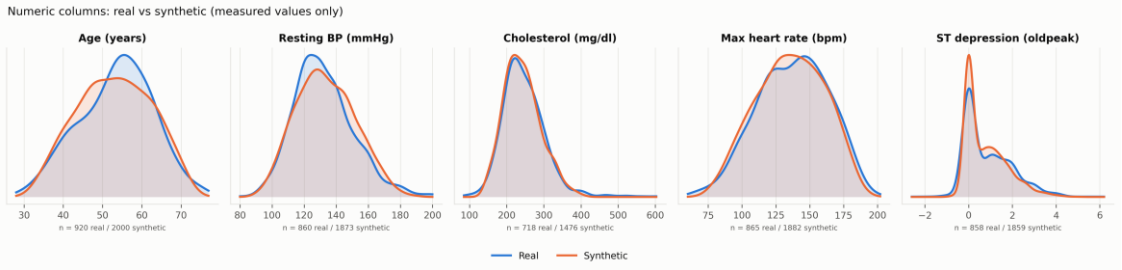


Figure 5.4: Distributions of the number columns: real (blue) vs MedTwin (orange), measured values only

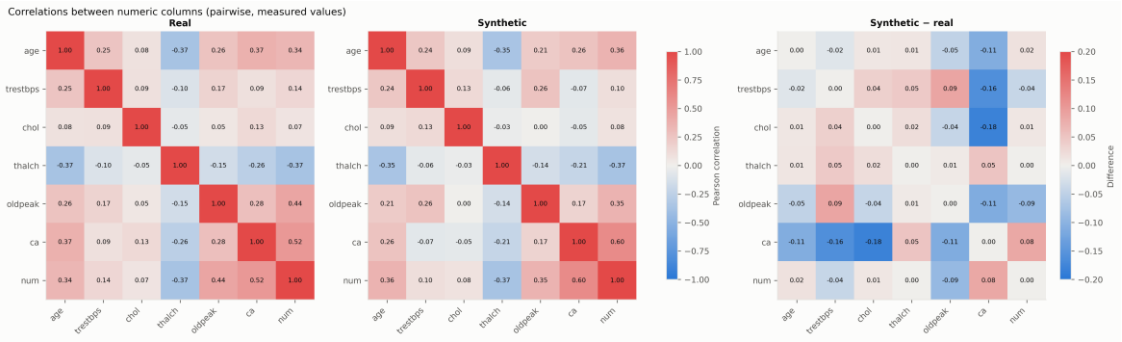


Figure 5.5: Correlations between number columns: real, MedTwin and their difference

5.8 Challenges Faced and Learnings

- **Training instability:** batches that happened to contain a constant column produced NaN values in the correlation loss, which corrupted the whole model. Skipping constant columns fixed this.
- **Hidden missing values:** zeros that really mean “not measured” (for example, Swiss cholesterol) distorted early results until they were handled explicitly through the settings file.
- **Fair evaluation:** early versions tested on patients the model had seen and reported a privacy score that actually measured realism. Holding out patients and separating detectability from privacy made the evaluation honest.
- **Statistical reliability:** single runs gave misleading differences between methods; repeating with five seeds and using paired confidence intervals showed which differences are real.
- **Long experiments:** the full comparison takes about an hour; saving each run's results and synthetic data allowed experiments to resume and new measures to be added without retraining.

CHAPTER 6
CONCLUSIONS & FUTURE SCOPE

6.1 Conclusions

This project designed, implemented and evaluated MedTwin, a generative AI system that produces realistic, useful and private synthetic patient records while keeping the real pattern of missing values. The key conclusions are:

1. Realistic synthetic patients, including their gaps

- **Outcome:** MedTwin produced the most realistic records of the four methods tested (detectability 61.5% on heart disease against 74.9–92.8%) and the highest fidelity on both datasets.
- **Significance:** treating "not measured" as part of the data, rather than filling it in, is what makes the records realistic — the ablation study shows the missing-value flags matter most.

2. Useful and private

- **Outcome:** a predictor trained on MedTwin's data keeps 97.5% (heart) and 98.3% (diabetes) of real-data performance, and privacy attacks found no memorisation.
- **Significance:** realism was gained without losing usefulness or privacy, unlike TabDDPM, which memorised its training patients on the diabetes data.

3. New ways to measure missing values

- **Outcome:** the new missing-value scores showed that MedTwin's gap patterns are almost as close to the real data as a second real sample (0.174 vs 0.152), and that only MedTwin places gaps at the right hospitals.
- **Significance:** these scores reveal differences that standard per-column checks hide, and can be used to evaluate any synthetic-data generator.

4. An honest privacy finding

- **Outcome:** keeping realistic gaps raises the membership attack on patients with rare patterns from 0.531 to 0.575, while overall privacy is unchanged.
- **Significance:** data owners should be aware of this small, consistent cost for patients with unusual gaps; it points to clear future work.

Limitations: both datasets are small (under 1000 patients); detectability is still above the ideal 50%; privacy was tested with attacks rather than guaranteed mathematically [27]; the

privacy audit of gaps needs enough rare-pattern patients and was inconclusive on diabetes; and the newest missing-value-aware generators were not yet included in the comparison.

6.2 Future Scope of Work

The work done in this project provides a solid base for further research. The following directions are recommended:

1. Comparison with missing-value-aware generators

- **Objective:** Compare MedTwin directly with the closest previous methods, such as MissDiff and the missing-pattern-preserving approach of Wang et al.
- **Applications:**
  - Benchmark all methods with the new missing-value scores.
  - Identify which design choices matter most for realistic gaps.
- **Significance:** Establishes where MedTwin stands among the newest methods and strengthens the research contribution.

2. Protecting patients with unusual gaps

- **Objective:** Reduce the privacy cost found for patients with rare missingness patterns.
- **Applications:**
  - Merge or coarsen rare patterns before release.
  - Train with differential privacy and measure the realism it costs.
- **Significance:** Allows realistic gaps to be kept while protecting the most exposed patients.

3. Generating data for new hospitals

- **Objective:** Use hospital conditioning to generate patients for a hospital the model has never seen.
- **Applications:**
  - Train on three hospitals and generate for the fourth.
  - Test whether models trained on these patients work at the new hospital.
- **Significance:** Supports data sharing and model development across hospitals.

4. Larger and time-based records

- **Objective:** Extend MedTwin to large hospital databases and records collected over time.
- **Applications:**

    - Patients with many visits and changing measurements.
    - Thousands of patients and many more columns.
- **Significance:** Moves the work closer to real-world hospital data and use.

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ANNEXURES

Annexure A: Sample synthetic patients

The first eight synthetic heart disease patients generated by MedTwin (seed 0). Blank values are shown as “not measured”. None of these records belongs to a real patient.

Table A.1: Sample of MedTwin's synthetic heart disease patients

age	gender	chest_pain	trestbps	chol	thalch	oldpeak	ca	thal	num	origin_study
58	Female	non-anginal	110	259	120	0.5	not measured	not measured	0	Hungary
60	Male	non-anginal	182	not measured	85	0	not measured	reversable defect	3	Switzerland
45	Male	asymptomatic	145	239	145	0	not measured	not measured	0	Hungary
42	Male	atypical angina	118	266	149	0	not measured	not measured	0	Hungary
45	Male	non-anginal	127	168	142	1.6	not measured	not measured	0	Hungary
51	Male	asymptomatic	143	317	149	2.1	not measured	not measured	2	Hungary
57	Male	asymptomatic	158	235	103	2.7	not measured	not measured	0	Hungary
61	Male	asymptomatic	172	196	116	not measured	not measured	not measured	1	VA Long Beach

Annexure B: Key code snippets

B.1 Creating the missing-value and zero flags (core/ingestion.py):

```
flags = pd.DataFrame(
    {missing_flag(c): df[c].isna().astype(float) for c in self.flagged_cols},
    index=df.index,
)
for col in self.zero_cols:
    flags[zero_flag(col)] = (df[col] == 0).astype(float)
```

B.2 Generator output layer (core/generator.py): sigmoid for numbers and flags, Gumbel-softmax for each category.

```
def forward(self, z, cond, hard=False):
    logits = self.model(torch.cat([z, cond], dim=1))
    outputs, start = [], 0
    for kind, width in self.output_info:
        segment = logits[:, start:start + width]
        if kind == "sigmoid":
            outputs.append(torch.sigmoid(segment))
        else:
            outputs.append(F.gumbel_softmax(segment, tau=GUMBEL_TAU, hard=hard))
        start += width
    return torch.cat(outputs, dim=1)
```

B.3 Pattern distance between real and synthetic gaps (core/missingness.py):

```
p_ref = patterns(reference, cols).value_counts(normalize=True)
p_oth = patterns(other, cols).value_counts(normalize=True)
union = p_ref.index.union(p_oth.index)
scores["pattern_tvd"] = float(
    0.5 * (p_ref.reindex(union, fill_value=0)
           - p_oth.reindex(union, fill_value=0)).abs().sum()
)
```

Annexure C: How to run the project

Task	Command
Install	pip install -r requirements.txt
Run the pipeline (heart disease)	python main.py
Run on diabetes	python main.py --config configs/diabetes.yaml
Dashboard	streamlit run app/dashboard.py
5-seed comparison	python research/multi_seed.py
Ablation study	python research/ablation.py
Sensitivity study	python research/ablation.py --study sensitivity
Figures	python research/visualizer.py --benchmark
Automated tests	python -m pytest tests -q

Annexure D: Project timeline

An early version of the project (April 2026) used 9 columns, filled missing values and tested on patients the model had seen. It was rebuilt to handle all columns with missing values and hospitals, to test on unseen patients and to compare against other methods. The main recorded steps since then are:

Date	Step
2026-09-29	Generated all 15 columns with “not measured” handling and hospital conditioning
2026-09-29	Added the oldpeak zero flag, 5-seed comparison and paper figures
2026-09-29	Moved dataset settings into configuration files; added the Pima diabetes dataset
2026-10-15	Added ablation switches and the ablation and sensitivity studies
2026-10-15	Separated detectability from privacy; added the correlation-gap measure
2026-10-15	Added the missing-value fidelity scores and privacy audit
2026-10-16	Verified all references; wrote and compiled the research paper